26.03.31

RLHF: R_θ : $(x, y_w > y_l)$ BTL (RM)

$$\mathbb{E}_{\pi_\theta}\left[R_\theta(x, \pi_\theta(x)) - \beta\, KL(\pi_\theta \| \pi_{SFT}) \right] \text{(PPO)}$$

maximize

DPO RM : R $\Rightarrow$ $\pi_\theta(R)$ (closed form)

$$R_\theta(\pi) = \beta \cdot \log \frac{\pi_\theta}{\pi_{SFT}} + \text{Constant.}$$

Fit.

$(x, y_w > y_l)$

$$\min_\theta \sum_{(x, y_w > y_l) \in D} -\log \sigma \left(\beta \cdot \log \frac{\pi_\theta(y_w)}{\pi_{SFT}(y_w)} - \beta \cdot \log \frac{\pi_\theta(y_l)}{\pi_{SFT}(y_l)} \right)$$

DPO: RL $\Rightarrow$ SL.